

A noncircular oval with convex unit-tangent iterates

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Abstract

For an oriented regular plane curve γ , define $\mathcal{T}\gamma = \gamma + \tau$, where τ is the positive unit tangent. We prove that there is a smooth noncircular strictly convex closed curve Γ such that $\mathcal{T}^n\Gamma$ is smooth and strictly convex for every $n \geq 0$, answering a question of Tabachnikov. The proof starts with a complete convex hairpin C such that $\mathcal{T}C$ is a translated copy of C . Periodizing its steering profile gives long closed convex models forming an approximate backward orbit with exponentially small L^1 curvature error. A regularizing inverse estimate shadows this approximate orbit by an exact infinite orbit. The initial curve is close to a long model of bounded width, so it cannot be a circle.

1 Introduction

The unit-tangent transform sends a circle of radius r to a circle of radius $\sqrt{r^2 + 1}$. We show that a noncircular oval can also remain convex under every iterate.

Let $\gamma : \mathbf{R}/L\mathbf{Z} \rightarrow \mathbf{R}^2$ be a smooth regular closed curve, parametrized by arclength, and let $\tau_\gamma = \gamma'$ be its unit tangent. The *unit-tangent transform* is the oriented curve

$$\mathcal{T}\gamma = \gamma + \tau_\gamma.$$

Curves are identified under orientation-preserving regular reparametrization, so $\mathcal{T}$ is independent of parametrization. Before each iterate, we reparametrize by arclength. Convex curves are oriented counterclockwise, with inward unit normal ν and $\tau' = k\nu$. A convex closed curve is the boundary of a convex body, traversed once. An *oval* is a smooth convex closed curve with $k > 0$ everywhere.

Tabachnikov asked whether an oval all of whose unit-tangent iterates are convex must be a circle [3, Section 9], [4, Conjecture 4]. We answer this question in the negative:

Theorem 1.1. *There is a noncircular oval Γ_0 such that $\mathcal{T}^n\Gamma_0$ is an oval for every integer $n \geq 0$.*

The construction uses long ovals of bounded transverse width. Their local model is a complete convex hairpin that $\mathcal{T}$ carries to a horizontal translate; see Figure 1.

In the bicycle model, R and $F = \mathcal{T}R$ are the rear and front tracks of a unit segment. The steering equation describes the angle between their tangents [2, Section 2]; see also [1, Proposition 2.6]. For a convex closed front of curvature less than one, this equation selects a unique convex rear on the branch where the steering angle lies between 0 and $\pi/2$. We denote this inverse by $\mathcal{B}$.

Sections 3–5 construct exact closed pairs $\mathcal{T}R_H = F_H$ and show that R_H and $F_{P(H)}$ have exponentially close intrinsic curvatures, where $P(H) = H - \Delta + o(1)$ and $\Delta > 0$. Section 6 turns these approximate inverse steps into an exact orbit. Section 7 excludes a circle by comparing width and perimeter.

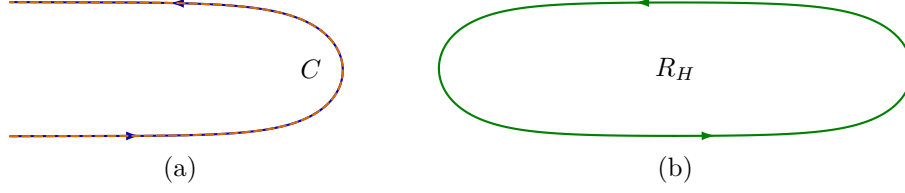


Figure 1: (a) A hairpin C and its unit-tangent image $C + \tau_C$. (b) The closed rear model R_H . The figure is illustrative and is not used in the proof.

2 The selected inverse

Suppose that $F = \mathcal{T}R$. Write s for front arclength and x for rear arclength. Let Θ and Ψ be the corresponding tangent angles, and put $\delta = \Theta - \Psi$. Resolving the front velocity in the rear Frenet frame gives

$$x_s = \cos \delta, \quad \Psi_s = \sin \delta.$$

Consequently, the front and rear curvatures satisfy

$$K = \delta_s + \sin \delta, \quad k = \tan \delta.$$

We write $\tau(\psi) = (\cos \psi, \sin \psi)$.

Geometric C^r convergence means C^r convergence after orientation-preserving parametrizations on a fixed circle. The selected inverse remains defined when the front curvature vanishes at some points.

Lemma 2.1. *Let F be a regular convex closed C^2 curve with curvature $0 \leq K \leq \kappa < 1$. There is a unique periodic solution of*

$$\delta_s = K - \sin \delta, \quad 0 \leq \delta \leq \arcsin \kappa.$$

It is strictly positive and defines a closed strictly convex rear $\mathcal{B}F$, with

$$0 < k_{\mathcal{B}F} \leq \frac{\kappa}{\sqrt{1 - \kappa^2}}. \quad (2.1)$$

The inverse preserves central symmetry, is continuous in geometric C^2 , and depends smoothly on smooth families of fronts. A C^r front, $r \geq 2$, has a C^{r+1} rear.

Proof. The steering vector field points into $I = [0, \arcsin \kappa]$ at both endpoints. Its period map therefore has a fixed point. The difference of two solutions in I satisfies

$$w_s = -a(s)w, \quad a(s) \geq \sqrt{1 - \kappa^2} > 0.$$

Periodicity forces $w = 0$. Any periodic solution on the larger branch $0 < \delta < \pi/2$ also lies in I : at a maximum of δ , the equation gives $\sin \delta \leq \kappa$.

Since $\delta_s + \delta = K + \delta - \sin \delta \geq K$, an integrating factor over a period of length L gives

$$(1 - e^{-L})\delta(s) \geq \int_{s-L}^s e^{t-s} K(t) dt > 0.$$

It is strict because $K \geq 0$ and $\int_0^L K = 2\pi$.

Choose a tangent-angle lift Θ for F , put $\Psi = \Theta - \delta$, and define

$$R(s) = F(s) - \tau(\Psi(s)).$$

Then $R_s = \cos \delta \tau(\Psi)$, so R is regular and closed and $\mathcal{T}R = F$. Its curvature is $\tan \delta > 0$, and its tangent turns through 2π ; hence it is embedded and bounds a strictly convex body. This proves (2.1). Uniqueness preserves central symmetry.

In normalized front arclength, the steering equation has a uniformly contracting period map near each front. Hence its fixed point δ depends continuously in C^1 on the front in C^2 . Since $x_s = \cos \delta > 0$, the rear and its arclength parametrization depend continuously in C^2 . The same argument gives smooth dependence on smooth parameters.

For regularity, a C^r front has $\Theta \in C^{r-1}$. The equation $\Psi_s = \sin(\Theta - \Psi)$ gives $\Psi \in C^r$, while $x_s = \cos(\Theta - \Psi) > 0$ gives a C^r change of arclength. Thus the rear tangent is C^r in rear arclength and the rear curve is C^{r+1} . $\square$

3 A translating hairpin

A *hairpin* is a complete embedded strictly convex curve whose tangent turns through π and whose ends are asymptotic to parallel lines. We construct a hairpin C such that $\mathcal{T}C = C + (V, 0)$.

Use tangent angle $0 < \theta < \pi$ as parameter, let ρ be the radius of curvature, and put

$$C_\theta = \rho(\theta)\tau(\theta), \quad f(\theta) = \rho(\theta) \sin \theta.$$

For $C = (X, Z)$, this gives

$$X' = f \cot \theta, \quad Z' = f. \tag{3.1}$$

The tangent angle of $C + \tau$ is $g(\theta) = \theta + d(\theta)$, where

$$d(\theta) = \arctan \frac{\sin \theta}{f(\theta)}.$$

Lemma 3.1. *Let $f \in C^\infty(0, \pi)$ satisfy $1 < m \leq f \leq M < \infty$. If*

$$\int_\theta^{\theta+d(\theta)} f(t) dt = \sin \theta, \quad d(\theta) = \arctan \frac{\sin \theta}{f(\theta)}, \tag{3.2}$$

then $g = \theta + d$ is an increasing diffeomorphism of $(0, \pi)$, and

$$C(\theta) + \tau(\theta) = C(g(\theta)) + (V, 0) \tag{3.3}$$

for a constant V .

Proof. Differentiation gives

$$f(g)g' = f + \cos \theta > 0.$$

Since $0 < d \leq \sin \theta / m < \pi - \theta$, the endpoint limits of g are 0 and π . Together with $g' > 0$, this makes g an increasing diffeomorphism. The integral identity gives the vertical component of (3.3).

For the horizontal component, use

$$\cot g = \frac{f \cos \theta - \sin^2 \theta}{\sin \theta (f + \cos \theta)}$$

to obtain

$$\frac{d}{d\theta} (X(\theta) + \cos \theta - X(g(\theta))) = f \cot \theta - \sin \theta - (f + \cos \theta) \cot g = 0. \quad \square$$

For a bounded measurable profile f with $1 < m \leq f \leq M$, set $A_f(t) = \int_0^t f(u) du$. For $0 < \theta < \pi$, define

$$D_f(\theta) = A_f^{-1}(A_f(\theta) + \sin \theta) - \theta.$$

It is well defined because $A_f(\pi) - A_f(\theta) \geq m(\pi - \theta) > \sin \theta$. Define

$$(\mathcal{P}f)(\theta) = \sin \theta \cot D_f(\theta).$$

Lemma 3.2. *The function $\mathcal{P}f$ is positive and continuous on $(0, \pi)$. Moreover, $\mathcal{P}$ is order preserving, and $f = \mathcal{P}f$ if and only if f satisfies (3.2).*

Proof. The primitive A_f and its inverse are continuous, and $0 < D_f \leq \sin \theta / m < \pi/2$, so $\mathcal{P}f$ is positive and continuous. If $f \leq h$, the integral of h reaches $\sin \theta$ no later than that of f ; hence $D_h \leq D_f$ and $\mathcal{P}f \leq \mathcal{P}h$. Finally, $f = \mathcal{P}f$ is equivalent, by the definition of D_f , to (3.2). $\square$

To construct a fixed point, define the barrier profiles

$$\begin{aligned} f_\varepsilon^-(\theta) &= \varepsilon^{-1} + \frac{2}{3}(1 - \cos \theta) - \varepsilon \cos \theta, \\ f_\varepsilon^+(\theta) &= \varepsilon^{-1} + \frac{2}{3}(1 - \cos \theta) + \varepsilon(2 + \cos \theta). \end{aligned}$$

For small $\varepsilon > 0$, both are bounded below by $\varepsilon^{-1} - \varepsilon > 1$, and $f_\varepsilon^+ - f_\varepsilon^- = 2\varepsilon(1 + \cos \theta) \geq 0$.

Lemma 3.3. *For all sufficiently small $\varepsilon > 0$,*

$$f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^+ \leq f_\varepsilon^+ \quad (0 < \theta < \pi). \quad (3.4)$$

Proof. For $d_f = \arctan(\sin \theta / f(\theta))$, put

$$\mathcal{R}(f) = \int_\theta^{\theta+d_f} f(t) dt - \sin \theta.$$

The integral increases with its upper limit, and cotangent decreases. Hence

$$\mathcal{R}(f) \geq 0 \iff D_f \leq d_f \iff \mathcal{P}f \geq f,$$

and the reversed inequalities are equivalent as well.

Each barrier has the form $f(\theta) = A + B \cos \theta$. With $d = d_f$, exact integration gives

$$\mathcal{R}(f) = f(\theta)d - \sin \theta + B \sin \theta (\cos d - 1) + B \cos \theta (\sin d - d).$$

Since $f(\theta) = \sin \theta \cot d$,

$$\frac{2\mathcal{R}(f)}{d^2 \sin \theta} = \frac{2(d \cot d - 1)}{d^2} + B \frac{2(\cos d - 1)}{d^2} + \frac{B \cos \theta}{f(\theta)} \frac{2(\sin d - d)}{d^2 \tan d}. \quad (3.5)$$

The three quotients extend to even analytic functions at $d = 0$, with values $-2/3$, -1 , and $-1/3$, respectively. For

$$f(\theta) = \varepsilon^{-1} + \frac{2}{3}(1 - \cos \theta) + \varepsilon(a + b \cos \theta),$$

we have $B = -2/3 + \varepsilon b$, $f(\theta)^{-1} = \varepsilon + O(\varepsilon^2)$, and $|d| \leq C\varepsilon \sin \theta$. Taylor's theorem gives

$$\begin{aligned} \frac{2\mathcal{R}(f)}{d^2 \sin \theta} &= -\frac{2}{3} - B - \frac{B \cos \theta}{3f(\theta)} + O(\varepsilon^2) \\ &= \varepsilon \left(-b + \frac{2}{9} \cos \theta \right) + O(\varepsilon^2), \end{aligned}$$

uniformly for $0 < \theta < \pi$. The analytic extensions of the quotients in (3.5) justify uniformity at both ends. For the lower barrier, $b = -1$, so the leading coefficient is at least $7/9$. For the upper barrier, $b = 1$, so it is at most $-7/9$. This proves the outer inequalities in (3.4); monotonicity gives the middle one. $\square$

Theorem 3.4. *For every sufficiently small $\varepsilon > 0$, equation (3.2) has a solution $f_\varepsilon \in C^\infty(0, \pi)$ with $f_\varepsilon^- \leq f_\varepsilon \leq f_\varepsilon^+$. The corresponding curve is a complete embedded strictly convex hairpin in a strip of finite width, and*

$$\mathcal{T}C = C + (V, 0), \quad V > 0.$$

Proof. Starting with $f_0 = f_\varepsilon^-$, define $f_{n+1} = \mathcal{P}f_n$. Lemma 3.3 and monotonicity give

$$f_\varepsilon^- \leq f_0 \leq f_1 \leq \cdots \leq f_\varepsilon^+.$$

Let $f = \lim_n f_n$ on $(0, \pi)$. The profiles share a lower bound $m > 1$, and monotone convergence gives $\int_0^\pi (f - f_n) \rightarrow 0$. Since $f_n \leq f$, $D_{f_n} \geq D_f$. Comparing the two defining integrals yields

$$0 \leq D_{f_n}(\theta) - D_f(\theta) \leq \frac{1}{m} \int_0^\pi (f - f_n)(t) dt \rightarrow 0.$$

Passing to the limit in $f_{n+1} = \mathcal{P}f_n$ gives $f = \mathcal{P}f$, hence the translator equation.

The fixed-point identity makes f continuous. Then A_f and its inverse are C^1 , so D_f and f are C^1 . Iterating this argument gives $f \in C^\infty(0, \pi)$.

Define C by (3.1). Its radius of curvature is $f/\sin \theta > 0$, and $Z' > 0$ makes it embedded. The bounds $m \leq f \leq M$ give infinite arclength at both ends, $X \rightarrow -\infty$, and finite one-sided limits of Z . Thus the ends are asymptotic to horizontal lines. Lemma 3.1 gives the translation law. At $\theta = \pi/2$,

$$V = - \int_{\pi/2}^{g(\pi/2)} f(t) \cot t dt > 0. \quad \square$$

Fix one such hairpin with curvature below $1/20$. All constants below may depend on this fixed choice and on any indicated derivative orders, but not on the period introduced in the next section.

Regard the hairpin as a rear track and its translate as the front. In front arclength s , let

$$y(s) = \sin \delta(s), \quad c(s) = \cos \delta(s) = \sqrt{1 - y(s)^2}.$$

Choose independent arclength origins so that both translated curves have the same intrinsic curvature K_* . If $x = x(s)$ is the corresponding rear arclength, then

$$x' = c, \quad K_*(s) = y + \frac{y'}{c}, \quad K_*(x(s)) = \frac{y}{c}.$$

Lemma 3.5. *There is $a > 0$ such that, for every $j \geq 0$,*

$$|y^{(j)}(s)| \leq C_j e^{-a|s|}, \quad |y^{(j)}(s)| \leq D_j y(s). \quad (3.6)$$

Moreover,

$$\int_{\mathbf{R}} y(s) ds = \pi, \quad \Delta := \int_{\mathbf{R}} (1 - c(s)) ds > 0,$$

and, for some $b_0 > 0$,

$$K_*(s) \geq b_0 y(s).$$

Proof. Let u be rear arclength, $\theta(u)$ its tangent angle, and $r(u) = \tan(\theta(u)/2)$. Then

$$(\log r)_u = \frac{\theta_u}{\sin \theta} = \frac{1}{f(\theta)}, \quad K_*(u) = \frac{\sin \theta(u)}{f(\theta(u))}.$$

The bounds $m \leq f \leq M$ give two-sided exponential bounds for $\sin \theta$ and K_* at both ends. For every $D > 0$, they also give the comparison

$$C_D^{-1} K_*(u) \leq K_*(u+h) \leq C_D K_*(u) \quad (|h| \leq D). \quad (3.7)$$

Here $r(u+h)/r(u)$ lies between $e^{-D/m}$ and $e^{D/m}$, so $2r/(1+r^2) = \sin \theta$ changes by a bounded factor. Also

$$\int_{\mathbf{R}} K_*(u)^2 du = \int_0^\pi \frac{\sin \theta}{f(\theta)} d\theta < \infty.$$

Let $\sigma(u)$ be front arclength at the corresponding point. Since $\sigma' = \sqrt{1 + K_*^2}$, the difference $\sigma(u) - u$ is bounded. The curvature formula for one tangent step gives

$$K_*(\sigma(u)) = \frac{K'_*(u) + K_*(u) + K_*(u)^3}{(1 + K_*(u)^2)^{3/2}}. \quad (3.8)$$

We prove by induction that

$$|K_*^{(j)}(u)| \leq C_j K_*(u) \quad (j \geq 0). \quad (3.9)$$

Equation (3.7) gives $K_* \circ \sigma = O(K_*)$, so (3.8) first gives $K'_* = O(K_*)$. Suppose the claim holds through order j . Derivatives of σ of orders 1 through $j+1$ are bounded. Differentiating $K_* \circ \sigma$ at most j times uses only derivatives of K_* through order j , evaluated at $\sigma(u)$, and bounded derivatives of σ . Thus each resulting term is $O(K_*(u))$. Differentiate

$$K'_* = (1 + K_*^2)^{3/2} (K_* \circ \sigma) - K_* - K_*^3$$

exactly j times. Every derivative of the prefactor is bounded, and every derivative of the other factors is $O(K_*)$. This proves the claim at order $j+1$. The exponential bounds for K_* therefore hold for all its derivatives.

The inverse correspondence $x = \sigma^{-1}$ has bounded derivatives of positive order, and

$$y(s) = \frac{K_*(x(s))}{\sqrt{1 + K_*(x(s))^2}}, \quad c(s) = \frac{1}{\sqrt{1 + K_*(x(s))^2}}.$$

The chain rule, (3.9), and $x(s) - s = O(1)$ give the derivative and tail bounds for y , with one common exponential rate $a > 0$.

The exact change of variables $y(s) ds = K_*(u) du$ gives the mass identity, since the rear tangent turns through π . The integral defining Δ is finite because $1 - c = O(y^2)$, and positive because $y > 0$. Finally, (3.7) gives

$$K_*(s) \geq b_0 K_*(x(s)) = b_0 \frac{y(s)}{c(s)} \geq b_0 y(s). \quad \square$$

4 Closed rear–front pairs

Periodize $y = \sin \delta$ with period $H > 0$:

$$Y_H(s) = \sum_{m \in \mathbf{Z}} y(s - mH). \quad (4.1)$$

For all sufficiently large H , $0 < Y_H < 1$. Define

$$\delta_H = \arcsin Y_H, \quad c_H = \sqrt{1 - Y_H^2},$$

and

$$K_H = Y_H + \frac{Y'_H}{c_H}, \quad k_H(x_H(s)) = \frac{Y_H(s)}{c_H(s)}, \quad x'_H = c_H. \quad (4.2)$$

Lemma 4.1. *Put $I_H = [-H/2, H/2]$. For all nonnegative integers r, q , there are $C_{r,q}, \beta_{r,q} > 0$ such that, for all sufficiently large H ,*

$$\|\partial_H^q(Y_H - y)\|_{C^r(I_H)} \leq C_{r,q} e^{-\beta_{r,q}H}. \quad (4.3)$$

Here s is fixed when H is differentiated. For fixed nonnegative integers j, k ,

$$\sum_{m \neq n} |y^{(j)}(s - mH)| |y^{(k)}(s - nH)| \leq C e^{-\beta H} Y_H(s) \quad (4.4)$$

for every $s \in \mathbf{R}$ and all sufficiently large H .

Proof. For $s \in I_H$ and $m \neq 0$,

$$|s - mH| \geq (|m| - 1/2)H, \quad \partial_H^q \partial_s^r y(s - mH) = (-m)^q y^{(r+q)}(s - mH).$$

The exponential bounds justify termwise differentiation and summation, and give (4.3).

The relative derivative bounds in (3.6) reduce (4.4) to $j = k = 0$. Write $n = m + \ell$. At least one of the arguments $s - mH$ and $s - nH$ has absolute value at least $|\ell|H/2$. Thus

$$y(s - mH)y(s - nH) \leq C e^{-a|\ell|H/2} (y(s - mH) + y(s - nH)).$$

Sum over m , then over $\ell \neq 0$. □

Lemma 4.2. *For $y_m(s) = y(s - mL)$ and all sufficiently large L ,*

$$\left| K_L(s) - \sum_m K_*(s - mL) \right| \leq C \sum_{m \neq n} y_m(s) y_n(s) \leq C e^{-\beta L} Y_L(s).$$

Proof. Put $G(z) = (1 - z^2)^{-1/2}$. The isolated and periodized formulas give

$$K_L - \sum_m K_*(s - mL) = \sum_m y'_m (G(Y_L) - G(y_m)).$$

For large L , all arguments of G lie in a fixed interval $[0, b]$, $b < 1$. On this interval G is Lipschitz, and $|y'_m| \leq D y_m$. Hence the last sum is bounded in absolute value by $C \sum_{m \neq n} y_m y_n$. Apply (4.4) for the second inequality. □

Proposition 4.3. *For all sufficiently large H , equations (4.1)–(4.2) define smooth centrally symmetric ovals R_H, F_H with $\mathcal{T}R_H = F_H$. Their perimeters are $2P(H)$ and $2H$, respectively, where*

$$P(H) = \int_0^H c_H(s) ds.$$

Moreover,

$$P(H) = H - \Delta + O(e^{-\beta H}), \quad P'(H) = 1 + O(e^{-\beta H}). \quad (4.5)$$

The curvatures of both curves are at most $\kappa_0 = 1/10$.

Proof. Choose Θ_H with $\Theta'_H = K_H$, and define F_H by $F'_H = \tau(\Theta_H)$. Since δ_H is H -periodic and $\int_0^H Y_H = \int_{\mathbf{R}} y = \pi$,

$$\int_0^H K_H ds = \int_0^H (\delta'_H + Y_H) ds = \pi.$$

The tangent reverses after each interval of length H , so F_H closes after two periods and is centrally symmetric. Set

$$\Psi_H = \Theta_H - \delta_H, \quad R_H = F_H - \tau(\Psi_H).$$

Then $\Psi'_H = Y_H$ and $R'_H = c_H \tau(\Psi_H)$, so R_H is the corresponding rear, also closed and centrally symmetric, with half-perimeter $P(H)$ and positive curvature k_H .

Lemma 4.2 and $K_* \geq b_0 y$ give

$$K_H \geq \sum_m K_*(s - mH) - Ce^{-\beta H} Y_H \geq (b_0 - Ce^{-\beta H}) Y_H > 0$$

for large H . Both closed curves have positive curvature and turning number one, hence are embedded ovals. On I_H , (4.3) also gives

$$K_H = K_* + O(e^{-\beta H}), \quad \frac{Y_H}{c_H} = \frac{y}{c} + O(e^{-\beta H}).$$

The isolated curvatures K_* and y/c are below $1/20$. Periodicity therefore bounds both model curvatures by $1/10$ for all sufficiently large H .

Put $\Phi(z) = 1 - \sqrt{1 - z^2}$. On the centered cell,

$$H - P(H) = \int_{-H/2}^{H/2} \Phi(Y_H(s)) ds.$$

By (4.3), smoothness of Φ on a fixed interval $[0, b]$, $b < 1$, and the omitted exponential tails,

$$H - P(H) = \int_{\mathbf{R}} \Phi(y(s)) ds + O(e^{-\beta H}) = \Delta + O(e^{-\beta H}).$$

Polynomial factors in H are absorbed by decreasing the positive exponential rate. Differentiating the centered integral at fixed s gives

$$\begin{aligned} \frac{d}{dH}(H - P(H)) &= \frac{1}{2}\Phi(Y_H(H/2)) + \frac{1}{2}\Phi(Y_H(-H/2)) \\ &\quad + \int_{-H/2}^{H/2} \Phi'(Y_H(s)) \partial_H Y_H(s) ds. \end{aligned}$$

Both endpoint terms are exponentially small. Since $|\Phi'(Y_H)| \leq CY_H$ and $\int_{I_H} Y_H = \pi$, the integral is bounded by $C\pi \|\partial_H Y_H\|_{L^\infty(I_H)}$. Equation (4.3) now proves (4.5). $\square$

Lemma 4.4. *Rotate F_H so that its tangent angle on I_H increases from 0 to π . Its vertical width is*

$$W_H = \int_{-H/2}^{H/2} \sin \Theta_H(s) ds.$$

There is a constant C such that $0 < W_H \leq C$ for all sufficiently large H .

Proof. Let Θ_* be the tangent angle of the isolated front hairpin, normalized by $\Theta_*(-\infty) = 0$. On I_H , the periodization estimates give

$$\|K_H - K_*\|_\infty \leq Ce^{-\beta H}, \quad \Theta_*(-H/2) = O(e^{-\beta H}).$$

With $\Theta_H(-H/2) = 0$, integration yields $\|\Theta_H - \Theta_*\|_\infty \leq CH e^{-\beta H}$. Hence

$$0 < W_H \leq \int_{\mathbf{R}} \sin \Theta_*(s) ds + CH^2 e^{-\beta H}.$$

The integral is the finite strip width of the fixed hairpin. □

5 Matching intrinsic curvatures

The curves R_H and $F_{P(H)}$ have the same perimeter. After suitable choices of arclength origin, the isolated rear and front, being translates, have the same intrinsic curvature K_* . Keep these origins fixed, and normalize the closed rear coordinate by $x_H(0) = x(0)$. For a period L , write

$$\overline{K}_L(u) = \sum_{j \in \mathbf{Z}} K_*(u - jL).$$

We compare both k_H and $K_{P(H)}$ with this periodization.

Theorem 5.1. *There are constants $C, \beta > 0$ such that, for all sufficiently large H ,*

$$\int_{\mathbf{R}/P(H)\mathbf{Z}} |k_H(u) - K_{P(H)}(u)| du \leq Ce^{-\beta H}. \quad (5.1)$$

Proof. Put $P = P(H)$, $I_H = [-H/2, H/2]$, and $J_H = x_H(I_H)$. The interval J_H has length P , so it translates by $P\mathbf{Z}$ tile the line. Our choice of origins aligns the central term K_* in both comparisons.

Exponential decay and the positive lower bounds for c_H and c give, for some $\alpha > 0$,

$$\begin{aligned} \|Y_H - y\|_{L^1(I_H)} + \|Y_H - y\|_{L^\infty(I_H)} &\leq Ce^{-\alpha H}, \\ \|c_H - c\|_{L^1(I_H)} + \|x_H - x\|_{L^\infty(I_H)} &\leq Ce^{-\alpha H}. \end{aligned} \quad (5.2)$$

The bound for $x_H - x$ follows by integrating $x'_H - x' = c_H - c$ and using $x_H(0) = x(0)$.

For the rear curvature, the exact identity

$$k_H(x_H(s)) dx_H = Y_H(s) ds$$

reduces the comparison to

$$\|k_H - \overline{K}_P\|_{L^1(J_H)} = \int_{I_H} |Y_H - c_H \overline{K}_P(x_H)| ds.$$

Using $y = cK_*(x)$, split the difference as

$$\begin{aligned} Y_H - c_H \bar{K}_P(x_H) &= (Y_H - y) + (cK_*(x) - c_H K_*(x_H)) \\ &\quad - c_H \sum_{j \neq 0} K_*(x_H - jP). \end{aligned}$$

The first two terms have exponentially small integrals by (5.2) and the boundedness of K_* and K'_* . Integrating the uniform coordinate error contributes a factor H , absorbed by decreasing α .

For the last term, change variables to rear arclength and unfold the periodization:

$$\begin{aligned} \int_{I_H} c_H \sum_{j \neq 0} K_*(x_H - jP) ds &= \int_{J_H} \sum_{j \neq 0} K_*(u - jP) du \\ &= \int_{\mathbf{R} \setminus J_H} K_*(u) du. \end{aligned}$$

On I_H , we have $x_H(s) = s + O(1)$, by (5.2) and the boundedness of $x(s) - s$. Thus the endpoints of J_H are $\pm H/2 + O(1)$, and the last integral is exponentially small. Hence

$$\|k_H - \bar{K}_P\|_{L^1(\mathbf{R}/P\mathbf{Z})} \leq Ce^{-\beta H}.$$

For the front, Lemma 4.2 gives

$$|K_P - \bar{K}_P| \leq Ce^{-\beta P} Y_P.$$

Integrating over a period and using $\int_0^P Y_P = \pi$ yields

$$\|K_P - \bar{K}_P\|_{L^1(\mathbf{R}/P\mathbf{Z})} \leq Ce^{-\beta P} \leq C'e^{-\beta' H}.$$

The triangle inequality proves (5.1). □

6 Backward shadowing

We obtain an exact infinite orbit by pulling back the model defects and summing the resulting normal variations. Curves in this section are centered, centrally symmetric, and oriented. Arclength origins are used only to compare model curvatures; they are not part of the topology on curves.

For a smooth path of curves, write its velocity as

$$\partial_t X = \xi\tau + \eta\nu.$$

The normal velocity η is unchanged by reparametrization. Define

$$W(\Gamma) = \int_0^1 \|\eta\|_{L^1} dt, \quad S_j(\Gamma) = \int_0^1 \|\partial_s^j \eta\|_{L^\infty} dt \quad (j = 0, 1, 2),$$

where s is arclength on the current curve and the norms are taken over its full perimeter. These quantities are additive under concatenation. We also allow paths that are piecewise smooth in t .

Lemma 6.1. *Let G_0, G_1 be centered, centrally symmetric ovals with the same half-perimeter L . Choose arclength origins whose tangent directions agree, and suppose their curvature functions satisfy $0 < \kappa^{(i)} \leq \kappa_*$. There is a path Γ from G_0 to G_1 , with curvature at most κ_* , such that*

$$W(\Gamma) + S_0(\Gamma) + S_1(\Gamma) \leq C(1 + L)^2 \|\kappa^{(1)} - \kappa^{(0)}\|_{L^1(0, L)}. \quad (6.1)$$

The constant C depends only on κ_* .

Proof. Set $\kappa_t = (1 - t)\kappa^{(0)} + t\kappa^{(1)}$, and let

$$\theta_t(s) = \theta_0 + \int_0^s \kappa_t(r) dr,$$

where θ_0 is the common initial tangent angle. Define

$$X_t(s) = \int_0^s \tau(\theta_t(r)) dr - \frac{1}{2} \int_0^L \tau(\theta_t(r)) dr \quad (0 \leq s \leq L),$$

and extend by $X_t(s + L) = -X_t(s)$. Each curvature function is L -periodic and has integral π , so this gives a smooth centered oval of curvature κ_t , parametrized by arclength.

Put $\epsilon = \|\kappa^{(1)} - \kappa^{(0)}\|_{L^1(0,L)}$. Then $\|\dot{\theta}_t\|_\infty \leq \epsilon$ and $\|\dot{X}_t\|_\infty \leq (3/2)L\epsilon$. Writing $\dot{X}_t = \xi\tau + \eta\nu$, the identity $\dot{\theta}_t = \eta_s + \kappa_t\xi$ gives

$$\|\eta\|_\infty \leq CL\epsilon, \quad \|\eta_s\|_\infty \leq C(1 + L)\epsilon.$$

The full perimeter is $2L$, so these estimates imply (6.1). $\square$

The inverse does not increase W and controls two derivatives of the rear normal velocity from one derivative of the front normal velocity.

Lemma 6.2. *Fix $0 < \hat{\kappa} < 1$. Let Γ be a smooth path of smooth regular convex closed fronts with curvature at most $\hat{\kappa}$. Their selected rears form a smooth path, and*

$$W(\mathcal{B}\Gamma) \leq W(\Gamma), \tag{6.2}$$

$$S_0(\mathcal{B}\Gamma) \leq C_0 W(\Gamma), \tag{6.3}$$

$$S_1(\mathcal{B}\Gamma) \leq C_1 (W(\Gamma) + S_0(\Gamma)), \tag{6.4}$$

$$S_2(\mathcal{B}\Gamma) \leq C_2 (W(\Gamma) + S_0(\Gamma) + S_1(\Gamma)). \tag{6.5}$$

The constants depend only on $\hat{\kappa}$.

Proof. Smooth dependence follows from Lemma 2.1. At a fixed time, let x be rear arclength and write $\dot{R} = \xi\tau + \eta_R\nu$. Since $F = R + \tau$,

$$\dot{\tau} = (\eta_{R,x} + k\xi)\nu, \quad \nu_F = -\sin \delta \tau + \cos \delta \nu.$$

Using $k = \tan \delta$, the tangential terms cancel and give

$$(1 + \partial_x)\eta_R = \sec \delta (\eta_F \circ s). \tag{6.6}$$

Here $s = s(x)$ is the corresponding front arclength. On $\mathbf{R}/\ell\mathbf{Z}$, the inverse of $1 + \partial_x$ is

$$(\mathcal{R}_\ell f)(x) = \frac{1}{1 - e^{-\ell}} \int_{x-\ell}^x e^{-(x-t)} f(t) dt.$$

Its kernel is positive and has integral one. Moreover,

$$\int |\sec \delta (\eta_F \circ s)| dx = \int |\eta_F| ds,$$

because $dx = \cos \delta ds$. This proves (6.2).

Lemma 2.1 bounds $\sec \delta$ and the rear curvature in terms of $\widehat{\kappa}$. Total turning gives

$$\ell \geq \frac{2\pi\sqrt{1-\widehat{\kappa}^2}}{\widehat{\kappa}} > 0.$$

Thus $\mathcal{R}_\ell$ maps L^1 to L^∞ with a uniform bound, proving (6.3). Equation (6.6) then gives (6.4). Differentiating it once yields

$$\eta_{R,xx} = (\sec \delta)_x (\eta_F \circ s) + \sec^2 \delta ((\eta_F)_s \circ s) - \eta_{R,x}.$$

The identity $\delta_x = \sec \delta (K - \sin \delta)$ bounds the coefficients using only $\widehat{\kappa}$, and proves (6.5). No bound on the front perimeter or on derivatives of its curvature is used. $\square$

Lemma 6.3. *Let G_j be regular closed curves joined by piecewise smooth paths Γ_j . Suppose all intermediate curvatures are bounded in absolute value by one constant, and*

$$\sum_j (S_0(\Gamma_j) + S_1(\Gamma_j) + S_2(\Gamma_j)) < \infty.$$

Then G_j converges in geometric C^2 to a regular closed C^2 curve.

Proof. Choose parametrizations that agree at the junctions, and concatenate the paths. On each segment, remove the tangential velocity by reparametrization. With $g = |X_u|$, the change of parameter solves

$$\partial_t \varphi = -\frac{\xi \circ \varphi}{g \circ \varphi}.$$

Use the endpoint parametrization to initialize the next segment. This leaves every S_j unchanged and gives $\partial_t X = \eta \nu$ in one transported periodic parameter. The inward-normal variation formulas are

$$\partial_t \tau = \eta_s \nu, \quad \partial_t \kappa = \eta_{ss} + \kappa^2 \eta, \quad \partial_t \log g = -\kappa \eta. \quad (6.7)$$

Summability makes X, τ, κ , and $\log g$ uniformly Cauchy. The limiting speed is positive because $\log g$ has finite total variation.

Switch to normalized arclength. On $[0, 1]$, put

$$\ell_j = \int_0^1 g_j(u) du, \quad h_j(u) = \ell_j^{-1} \int_0^u g_j(v) dv, \quad Y_j = X_j \circ h_j^{-1}.$$

The maps h_j converge in C^1 to a map with positive derivative. Consequently,

$$Y'_j = \ell_j \tau_j \circ h_j^{-1}, \quad Y''_j = \ell_j^2 (\kappa_j \nu_j) \circ h_j^{-1}$$

converge uniformly, as does Y_j . Their limit is a regular closed C^2 parametrization. $\square$

Theorem 6.4. *Let Q_n be centered, centrally symmetric ovals, and put $A_n = \mathcal{B}Q_{n+1}$. Suppose Q_n and A_n have the same half-perimeter L_n , and all these curves have curvature at most one constant $\kappa_0 < 1$. For each pair, choose arclength origins with the same tangent direction, and define*

$$e_n = (1 + L_n)^2 \|\kappa_{Q_n} - \kappa_{A_n}\|_{L^1(0, L_n)}, \quad r_n = \sum_{m=n}^{\infty} e_m.$$

There are constants $\eta_, C > 0$, depending only on κ_0 , with the following property. If $r_0 \leq \eta_*$, all backward iterates*

$$Z_n^{(N)} = \mathcal{B}^{N-n} Q_N \quad (N \geq n)$$

are defined. For each fixed n , they converge as $N \rightarrow \infty$ in geometric C^2 to smooth ovals X_n satisfying

$$X_n = \mathcal{B}X_{n+1}, \quad \mathcal{T}X_n = X_{n+1},$$

and

$$d_H(X_n, Q_n) + |\text{Per}(X_n) - 2L_n| \leq Cr_n. \quad (6.8)$$

Here d_H denotes Hausdorff distance.

Proof. Defect paths. By Lemma 6.1, there are paths $\Lambda_n : Q_n \rightsquigarrow A_n$ with curvature at most κ_0 and

$$W(\Lambda_n) + S_0(\Lambda_n) + S_1(\Lambda_n) \leq d_n, \quad d_n = C_{\text{int}} e_n.$$

Put $a_n = \sum_{m \geq n} d_m = C_{\text{int}} r_n$. All constants below depend only on κ_0 .

Set $\hat{\kappa} = (1 + \kappa_0)/2$, and take the constants C_0, C_1, C_2 from Lemma 6.2 for this curvature bound.

Put

$$M_0 = \max\{1, C_0\}, \quad M_1 = \max\{1, C_1(1 + M_0)\},$$

and

$$C_* = C_2(1 + M_0 + M_1) + \hat{\kappa}^2 C_0.$$

Choose $\eta_* > 0$ so small that

$$C_* C_{\text{int}} \eta_* < \hat{\kappa} - \kappa_0.$$

Invariant neighborhoods. Let $\mathcal{D}_n$ consist of the curves joined to Q_n by a piecewise smooth path of centered, centrally symmetric ovals with curvature at most $\hat{\kappa}$ and

$$W \leq a_n, \quad S_0 \leq M_0 a_n, \quad S_1 \leq M_1 a_n.$$

We claim

$$\mathcal{B}(\mathcal{D}_{n+1}) \subseteq \mathcal{D}_n. \quad (6.9)$$

Let Γ be an admissible path from Q_{n+1} to $C \in \mathcal{D}_{n+1}$. Its inverse path exists by Lemma 2.1 and starts at A_n . Lemma 6.2 gives

$$W(\mathcal{B}\Gamma) \leq a_{n+1}, \quad S_0(\mathcal{B}\Gamma) \leq C_0 a_{n+1},$$

$$S_1(\mathcal{B}\Gamma) \leq C_1(1 + M_0) a_{n+1}, \quad S_2(\mathcal{B}\Gamma) + \hat{\kappa}^2 S_0(\mathcal{B}\Gamma) \leq C_* a_{n+1}.$$

The inverse path starts with curvature at most κ_0 . Suppose its maximum curvature first reaches $\hat{\kappa}$ at time t . Equation (6.7) then gives

$$\|\kappa(t) - \kappa(0)\|_{\infty} \leq S_2(\mathcal{B}\Gamma) + \hat{\kappa}^2 S_0(\mathcal{B}\Gamma) \leq C_* a_{n+1} < \hat{\kappa} - \kappa_0.$$

This contradicts the choice of t . Concatenate Λ_n with $\mathcal{B}\Gamma$. Since $a_n = d_n + a_{n+1}$, the resulting path satisfies the bounds defining $\mathcal{D}_n$, proving (6.9). Hence every finite backward iterate is defined and lies in its corresponding neighborhood.

Convergence. Every curve on Λ_N belongs to $\mathcal{D}_N$: the path up to that curve satisfies the defining bounds because $d_N \leq a_N$. Repeated use of (6.9) therefore shows that

$$\Gamma_{n,N} = \mathcal{B}^{N-n}\Lambda_N$$

exists and stays in $\mathcal{D}_n$. It joins $Z_n^{(N)}$ to $Z_n^{(N+1)}$. The inverse estimates, applied repeatedly, give

$$W(\Gamma_{n,N}) \leq d_N, \quad S_0(\Gamma_{n,N}) \leq M_0 d_N, \quad S_1(\Gamma_{n,N}) \leq M_1 d_N.$$

For $N \geq n+1$, at least one inverse has been applied, so

$$S_2(\Gamma_{n,N}) \leq C_2(1 + M_0 + M_1)d_N.$$

For fixed n , Lemma 6.3 therefore applies to the sequence with $N \geq n+1$. Let X_n be its geometric C^2 limit.

The limit has nonnegative curvature. Its unit tangent has degree one, since the unit tangent maps converge uniformly. Fix a point and project the curve onto the inward normal there. During one circuit the projection increases, then decreases to its initial value. It can equal that value only on the initial or terminal straight portion; there the curve moves along the tangent and cannot return early. Hence the curve is embedded and lies on the inward side of every tangent line. It bounds a convex body and has curvature at most $\hat{\kappa} < 1$.

By the continuous dependence in Lemma 2.1, the identities $Z_n^{(N)} = \mathcal{B}Z_{n+1}^{(N)}$ pass to the limit:

$$X_n = \mathcal{B}X_{n+1}.$$

The same lemma gives positive rear curvature and gains one derivative. Applying it repeatedly to $X_n = \mathcal{B}^r X_{n+r}$ makes every X_n smooth. The definition of $\mathcal{B}$ then gives the forward identity.

Distance from the models. Every $Z_n^{(N)}$ can be joined to Q_n by a defining path for $\mathcal{D}_n$. Normal parametrization bounds the Hausdorff distance by S_0 , while the first variation of perimeter gives $|\dot{\ell}| \leq \hat{\kappa}\|\eta\|_{L^1}$. Consequently,

$$d_H(Z_n^{(N)}, Q_n) \leq M_0 a_n, \quad |\text{Per}(Z_n^{(N)}) - 2L_n| \leq \hat{\kappa} a_n.$$

Passing to the limit and using $a_n = C_{\text{int}} r_n$ proves (6.8). $\square$

If $r_0 < \infty$, then $r_n \rightarrow 0$, so the theorem applies after discarding finitely many initial indices.

7 Proof of the main theorem

Proof of Theorem 1.1. Use the fixed translating hairpin from Section 3. For large H , Proposition 4.3 gives

$$P'(H) > \frac{1}{2}, \quad \frac{\Delta}{2} \leq H - P(H) \leq \frac{3\Delta}{2}.$$

Hence each sufficiently large H_0 determines a unique sequence with

$$P(H_{n+1}) = H_n, \quad H_n \geq H_0 + \frac{\Delta}{2}n. \quad (7.1)$$

Choose centered representatives of the exact pairs $\mathcal{T}R_{H_n} = F_{H_n}$, and set $Q_0 = F_{H_0}$. Inductively rotate the whole pair $R_{H_{n+1}}, F_{H_{n+1}}$ so that the rear tangent and the tangent of Q_n agree at $u = 0$ in the intrinsic coordinates of Theorem 5.1. Define

$$Q_{n+1} = F_{H_{n+1}}, \quad A_n = \mathcal{B}Q_{n+1} = R_{H_{n+1}}.$$

Because $\mathcal{T}$ commutes with rotations, these choices preserve the exact pairs and all intrinsic curvature estimates. By (7.1), Q_n and A_n have the same half-perimeter H_n , and their chosen tangent directions agree.

By Theorem 5.1, the defects in Theorem 6.4 satisfy

$$e_n \leq C(1 + H_n)^2 e^{-\beta H_{n+1}}.$$

Absorbing the polynomial factor into the exponential and using (7.1), we obtain, for some $\beta' > 0$,

$$r_0 \leq C \sum_{n=0}^{\infty} e^{-\beta'(H_0 + n\Delta/2)} \leq C' e^{-\beta' H_0} \rightarrow 0 \quad (H_0 \rightarrow \infty). \quad (7.2)$$

All model curvatures are at most $\kappa_0 = 1/10$. For sufficiently large H_0 , Theorem 6.4 gives smooth ovals X_n with $\mathcal{T}X_n = X_{n+1}$ and

$$d_H(X_0, Q_0) + |\text{Per}(X_0) - 2H_0| \leq Cr_0. \quad (7.3)$$

By Lemma 4.4, Q_0 has uniformly bounded width in one direction. A Hausdorff perturbation of size d changes directional width by at most $2d$, so X_0 also has bounded width in that direction. On the other hand, (7.2) and (7.3) give $\text{Per}(X_0) = 2H_0 + o(1) \rightarrow \infty$. Since a circle has perimeter π times its width in every direction, X_0 is noncircular for large H_0 . Take $\Gamma_0 = X_0$. $\square$

References

- [1] G. Bor, M. Levi, R. Perline, and S. Tabachnikov, Tire tracks and integrable curve evolution, *International Mathematics Research Notices* **2020**, no. 9, 2698–2768. doi:10.1093/imrn/rny087.
- [2] M. Levi and S. Tabachnikov, On bicycle tire tracks geometry, hatchet planimeter, Menzin’s conjecture, and oscillation of unicycle tracks, *Experimental Mathematics* **18** (2009), no. 2, 173–186. doi:10.1080/10586458.2009.10128894.
- [3] S. Tabachnikov, A baker’s dozen of problems, *Arnold Mathematical Journal* **1** (2015), no. 1, 59–67. doi:10.1007/s40598-014-0001-3.
- [4] S. Tabachnikov, Iterating skew evolutes and skew involutes: a linear analog of bicycle kinematics, *Moscow Mathematical Journal* **23** (2023), no. 4, 625–640. doi:10.17323/1609-4514-2023-23-4-625-640.